



Mitsubishi Chemical & AIST: GIC 2026

Advanced Materials: Leverage AI-enhanced quantum eigensolvers to accelerate the simulation and discovery of advanced semiconductor and chemical materials.



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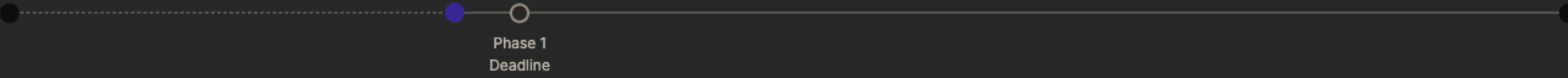
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Mitsubishi Chemical & The National Institute of Advanced Industrial Science and Technology (AIST)

Harnessing the Generative Quantum Eigensolver for Next-Generation Materials Design

Advanced Materials

Advanced materials design drives innovation across the chemical and semiconductor industries, yet conventional computational approaches face limitations in accuracy and scalability when exploring complex molecular and material systems. This challenge centers on the Generative Quantum Eigensolver (GQE), an AI-driven quantum application that combines generative machine learning models with quantum eigensolvers to enable more accurate quantum simulations and efficient exploration of vast materials design spaces. Participants will investigate approaches for applying GQE within a Quantum Materials Informatics platform to improve quantum simulation accuracy for material properties, efficiently generate molecular and structural candidates, and accelerate materials discovery beyond the capabilities of classical simulation methods, including the simulation of extreme ultraviolet semiconductor materials.

Phase 1

Phase 1 focuses on three things: building your team, selecting your Challenge track, and developing your initial concept.

Team Composition

The Global Industry Challenge is a collaborative competition; teams can be formed locally or internationally with between 1 and 5 teammates.

Find teammates here.

Join a team here.

Teams may have between 1 and 5 participants. We recommend 3 to 5 members to balance the workload effectively. Strong teams are diverse in skill. We encourage you to fill the following roles:

- Coder
- Data / Technical Lead
- Content / Domain Expert
- Business / Commercial Lead
- Project Manager / Team Coordinator

What do I submit for Phase 1?

Create a 1-page (plus cover page) maximum PDF, using 11-point Times New Roman font and single spacing, and submit it through the relevant Aqora Competition.

File Format Requirement: TeamName__Phase1_VersionX.pdf

Include the following details:

- Cover Page (not included in page count) [GIC 2026 Cover Page.docx](#)
- Team Qualifications for Tackling the Challenge (Education, Job Title and Organization)
- Brief Description of the Steps You Will Use to Solve this Challenge
- High-level overview of your proposed solution
- Technical approach, including how you will utilize quantum computing feasibility and advantage over classical methods
- Projected industry impact

First Phase Submission Deadline: Sunday, April 19, 11:59 PM (EST).

Judging Criteria

Teams will be evaluated on two dimensions. First, technical proficiency: judges will assess the depth and quality of your team's quantum computing knowledge and execution capability. Second, a higher weight is placed on conceptual feasibility: your Phase 1 submission must define a credible, technically grounded approach that benchmarks quantum methods against classical baselines and articulates measurable industry impact.

Full Challenge Description:

Advanced materials design is a critical driver of innovation across the chemical and semiconductor industries, yet conventional computational approaches face fundamental limitations in accuracy and scalability when exploring complex molecular and material systems. Under Japan's Strategic Innovation Promotion Program (SIP), Mitsubishi Chemical is leading an industry-driven effort to integrate artificial intelligence with quantum computing to overcome these barriers.

This challenge centers on the Generative Quantum Eigensolver (GQE)—an AI-driven quantum application jointly developed with AIST/G-QuAT, the University of Toronto, and NVIDIA. GQE combines generative machine learning models with quantum eigensolvers to enable more accurate quantum simulations and efficient exploration of vast materials design spaces. Participants will explore how GQE can be leveraged within a Quantum Materials Informatics (MI) Platform to accelerate materials discovery beyond the capabilities of classical simulation methods.

Your challenge is to investigate and propose approaches for applying GQE to materials informatics workflows, focusing on areas such as improving quantum simulation accuracy for material properties, efficiently generating molecular and structural candidates, and scaling exploration across complex chemical design spaces. As a concrete industrial use case, the challenge highlights the simulation of extreme ultraviolet (EUV) semiconductor materials, where high precision at the quantum level is essential.

Through this challenge, participants will gain hands-on exposure to cutting-edge quantum–AI integration and contribute ideas toward next-generation materials development with real-world industrial impact.